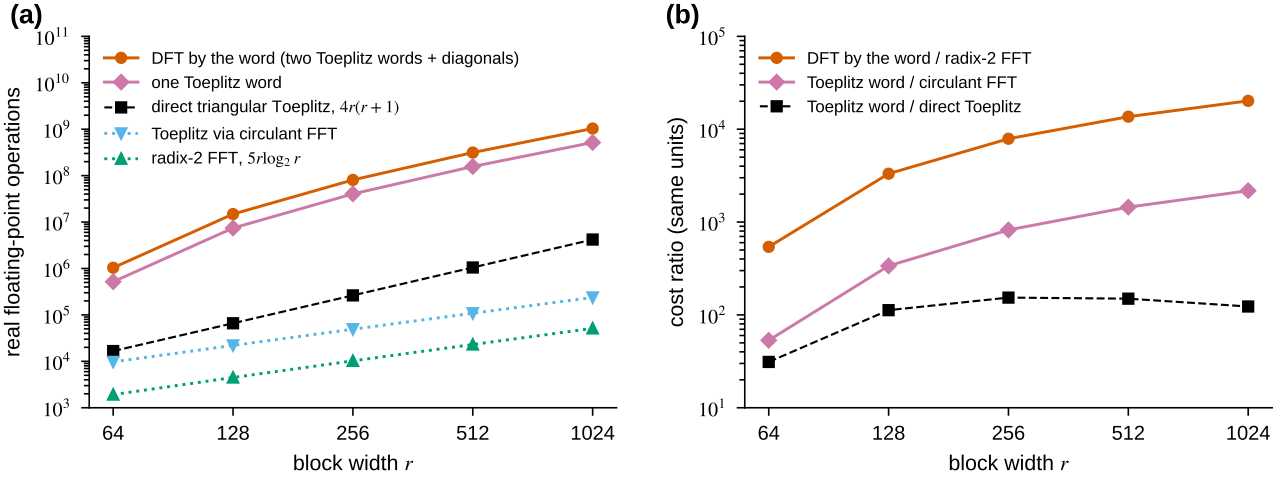
(a)
real floating-point operations
10^{11}
10^{10}
10^{9}
10^{8}
10^{7}
10^{6}
10^{5}
10^{4}
10^{3}
DFT by the word (two Toeplitz words + diagonals)
one Toeplitz word
direct triangular Toeplitz, $4r(r+1)$
Toeplitz via circulant FFT
radix-2 FFT, $5r\log_2 r$
64
128
256
512
1024
block width r
(b)
cost ratio (same units)
10^{5}
10^{4}
10^{3}
10^{2}
10^{1}
DFT by the word / radix-2 FFT
Toeplitz word / circulant FFT
Toeplitz word / direct Toeplitz
64
128
256
512
1024
block width r